Primordial Timing Function: Net-Zero Displacement Proof, Pi-Digit Epoch Clock, and Epoch-5 Boundary

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PAPER_1153 – Primordial Timing Function: Net-Zero Displacement Proof, Pi-Digit Epoch Clock, and Epoch-5 Boundary

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$$\int_0^1 \cos(\pi t_n)\, dt_n = 0 \qquad D_A + D_B = +3 + (-3) = 0$$

Abstract

This paper registers the Primordial Timing Function (PTF) as PAPER_1153, proving that the UQFF cosmic temporal cycle has **zero net displacement** across a complete forward-backward sequence. The PTF encodes the canonical π-digit epoch clock, identifies the five cosmic epochs, and places Epoch 5 at the **reverse boundary** (not-yet-predicted). The net-zero proof unifies three independent proofs: (1) integer arithmetic on forward/backward step counts, (2) the $\cos(\pi t_n)$ integral, and (3) Fibonacci-structure forward/backward ratio.

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1. PTF Definition

1.1 Basic Structure

The Primordial Timing Function (PTF) governs the oscillation of $\cos(\pi t_n)$ across one complete simulation cycle:

$$t_n \in [0, 1], \quad f_{\mathrm{TRZ}}(t_n) = \cos(\pi t_n)$$

One cosmic cycle consists of: - **Forward phase:** $f = 3$ steps (Fibonacci $F_4 = 3$) - **Backward phase:** $b = 2$ steps (Fibonacci $F_3 = 2$) - **Repeats:** $n = 3$ cycles ($= \lfloor \pi \rfloor$)

1.2 Fibonacci Structure

The forward/backward step counts are consecutive Fibonacci numbers:

$$f = F_4 = 3, \quad b = F_3 = 2, \quad \frac{f}{b} = \frac{3}{2} = \phi^2 - 1 \approx 1.618 - 1$$

This ratio $f/b = 3/2$ is the **golden-ratio residual** governing UQFF temporal asymmetry.

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2. Net-Zero Displacement: Three Independent Proofs

2.1 Proof I: Integer Arithmetic

Define net displacements:

$$D_A = +3 \quad \text{(forward: 3 steps, positive)}, \qquad D_B = -3 \quad \text{(backward: 2 steps} \times (-3/2) = -3)$$

Net displacement per cycle:

$$D_{\text{net}} = D_A + D_B = +3 + (-3) = 0 \quad \checkmark$$

Three complete repetitions: $n = 3 \Rightarrow \sum 0 = 0$.

2.2 Proof II: Integral of the Oscillation Function

The $\cos(\pi t_n)$ integral over one complete normalised period is exactly zero:

$$\int_0^1 \cos(\pi t_n)\, dt_n = \left[\frac{\sin(\pi t_n)}{\pi}\right]_0^1 = \frac{\sin(\pi) - \sin(0)}{\pi} = \frac{0-0}{\pi} = 0 \quad \checkmark$$

This is a **closed loop** in the functional space: the oscillation returns to its starting state with zero net accumulation.

2.3 Proof III: CPT Symmetry

The CPT-symmetric pair structure:

$$\{t, t_n, f_{\text{TRZ}}\}_{\text{fwd}} = \{t, t_n, f_{\text{TRZ}}\}^*_{\text{bwd}} \quad \text{(charge-parity-time conjugate)}$$

At $t < 0$: $t_n \to 1 - t_n$, $f_{\text{TRZ}} \to -f_{\text{TRZ}}$. The integral of f_{TRZ} over one complete CPT pair: $\int_0^1 + \int_1^0 = 0$.

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3. Pi-Digit Epoch Clock

3.1 Epoch Assignment

The digits of $\pi = 3.14159265358979\ldots$ encode the duration of each cosmic epoch. The epoch clock reads the decimal expansion in triplets:

Epoch	Pi Digits	Epoch Characteristic
1	$[3, 1, 4]$	Formation: 3-1-4 (forward-stop-backward)
2	$[1, 5, 9]$	Expansion: 1-5-9 (seed-burst-accumulation)
3	$[2, 6, 5]$	Stabilisation: 2-6-5 (pair-hexagon-pentagon)
4	$[3, 5, 8]$	Acceleration: 3-5-8 (Fibonacci triple)
5	$[9, 7, 9]$	**NOT YET PREDICTED** (reverse boundary)

3.2 Epoch 5: The Reverse Boundary

Epoch 5 ($[9, 7, 9]$) is the **reverse boundary** of the cosmic cycle: - 9 = maximum single digit ($= 10 - 1$, the final decimal state) - 7 = prime boundary ($\pi(7) = 4$, fourth prime threshold) - 9 = return to maximum (closed boundary)

The simulation engine does not project Epoch 5 forward from known physics; it is identified as the point where the backward phase ($D_B = -3$) begins.

3.3 Forward Phase (Epochs 1–4) and Backward Phase (Epoch 5)

$$\underbrace{[3, 1, 4], [1, 5, 9], [2, 6, 5], [3, 5, 8]}_{\text{forward: } D_A = +3} \qquad \underbrace{[9, 7, 9]}_{\text{backward: } D_B = -3}$$

The boundary between Epoch 4 and Epoch 5 is the **temporal inversion point** where t_n reverses sign.

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4. Constant Table

Symbol	Value	Physical Meaning
f	3	Forward steps per half-cycle = F_4
b	2	Backward steps per half-cycle = F_3
n	3	Repetitions = $\lfloor \pi \rfloor$
D_A	$+3$	Forward net displacement
D_B	-3	Backward net displacement
D_{net}	0	Closed-loop condition
f/b	$3/2$	Golden residual
π_5	$[9, 7, 9]$	Epoch 5 digit triplet

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5. Physical Consequences

5.1 SSq Drift is Convergent

Because $D_{\text{net}} = 0$, the [SSq] drift (+0.000001 per tick) does not accumulate unboundedly. After one complete epoch cycle, the drift is absorbed by the $\kappa = 0.0005$ gradient-descent restoring force:

$$[\text{SSq}](T) = 0.57 + \delta - \kappa\delta T \to 0.57 \quad \text{as } T \to \infty$$

5.2 Closed-Loop Causality

The PTF net-zero condition enforces closed-loop causality: every forward-phase physical event has an exact CPT-conjugate backward-phase event. This is the UQFF implementation of the Wheeler-DeWitt constraint $\hat{H}|\Psi\rangle = 0$.

5.3 Functional PTF Validation

The simulation engine's `validate_primordial_timing_function()` tests: - $|D_A + D_B| < \epsilon$ - $\left|\int_0^1 \cos(\pi t_n) dt_n\right| < 10^{-10}$ - Fibonacci ratio: $|f/b - 3/2| < \epsilon$ - Epoch count: exactly 5 epochs from π decimal expansion

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6. Conclusions

The Primordial Timing Function establishes: 1. **Net-zero displacement** is proven by three independent methods 2. The **pi-digit epoch clock** encodes cosmic epoch durations in $\pi = 3.14159\ldots$ 3. **Epoch 5** $[9, 7, 9]$ is the reverse boundary — not predictable from forward-phase physics 4. CPT symmetry and Wheeler-DeWitt closure are unified in the PTF framework

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